DRAFT, not for distribution. Readable version generated from the LaTeX sources (tools/tex_to_html.py); the layout differs from the VJCS template.

Answer, Clarify or Refuse? Rules, a Reranker and a Small LLM on Real Citizen Questions about Commune-Level Administrative Procedures in Vietnam

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Abstract. A commune- and ward-level procedure assistant must decide whether to answer, clarify, or refuse. We evaluate it on real questions from two Vietnamese public-service portals, with test sets locked before the code and one run per configuration. On one frozen system, behavior accuracy falls from 77.2% on team-written to 56.5% on real questions, 79% of which must be refused. A transparent rule layer raises balanced accuracy from 0.647 to 0.814 and end-to-end answers from 19 to 35 of 59, and on a second locked set raises correct refusals from 175 to 297 of 335. A reranker and a local 4-billion-parameter language model do not significantly improve its balanced or end-to-end accuracy, nor do gates learned *post hoc*; that model deciding over retrieved procedures is no better than always refusing, and a second rule round tied the first on new questions. On random traffic, most answers of every configuration are wrong.
Keywords: Administrative procedures; question answering; abstention; Vietnamese; rule-based systems; evaluation.

1. Introduction

Many everyday administrative procedures in Vietnam are handled by commune-level offices: civil status, residence, identity cards, certified copies, social benefits. The National Public Service Portal publishes the record of each procedure[1], but citizens rarely ask in its language. They write letters with a greeting, a family story and abbreviations, often without diacritics, and many ask not how to carry out a procedure but about a file already submitted, a complaint, a fine, or the benefits of police officers. Answering these from procedure records gives confident but wrong guidance, which costs citizens time and trust[2,3].

The task is therefore a three-way decision before it is a retrieval problem: *answer* with one procedure, *clarify* when several fit, or *refuse*. Abstention is a known weak point of language models[4,5] and intent classifiers struggle with out-of-scope queries[6], yet prior work on Vietnamese public-service question answering evaluates only answerable questions[7].

This paper is an evaluation study of that decision on real citizen questions, for an assistant that covers only procedures handled at the commune level (communes, wards and special zones), not all administrative procedures in Vietnam, and that runs on one local machine so that questions need not leave the office[8]. Test sets are locked before the code they evaluate, and every configuration runs once and is reported. We find that:

- our team-written test set overstated performance: 79% of real questions are out of scope (88–92% in random samples), against 11%, and top-1 accuracy falls from 71.9% to 33.9%;
- a transparent rule layer closes much of the gap on two locked test sets, through an out-of-scope gate (refusals) and a citizen lexicon (answers) according to *post hoc* ablations, with no net change on the team-written set;
- a reranker, a local 4-billion-parameter large language model (LLM) and gates learned *post hoc* do not significantly improve balanced or end-to-end accuracy, and letting the model decide over retrieved procedures is no better than always refusing in balanced accuracy;
- a second round of rules that looked better on development ties the first on new questions, and on random traffic

no configuration has more correct than wrong answers.

2. Related Work

Vietnamese public-service and legal question answering. Pham Duy and Le Thanh retrieve legal passages for public-service questions from the portal's pages of frequently asked questions with BM25+, a contrastive PhoBERT encoder and a cross-encoder, reporting retrieval metrics[7]. ViGPTQA trains Vietnamese instruction-following LLMs for question answering, including law[9], and Hien *et al.* combine vector retrieval over 45,000 legal documents with an LLM[10]; their system routes out-of-scope questions to experts, but this step is not evaluated. None measures refusal on real citizen questions.

Abstention and out-of-scope detection. Selective classification trades coverage for risk[11]; unanswerable questions caused a large drop for strong SQuAD readers[12]; a trained calibrator helps a reader abstain under domain shift[5]; on CLINC150, intent classifiers that do well in scope struggle to flag out-of-scope queries[6]. A survey covers abstention in LLMs[4], and clarifying questions are a third option between answering and refusing[13].

Who decides. In retrieval-augmented generation (RAG)[14], Self-RAG and corrective RAG, the model or a retrieval evaluator decides what to do[15,16]; programmable guardrails let developer-written rails decide instead[17]. In e-government, a governance pipeline escalates to humans[2], and a benchmark finds that answering correctly does not imply acknowledging what one does not know[3]. LLMs can outperform crowd workers at text annotation[18].

3. Task and Data

Task. The system outputs *answer* with a procedure identifier, *clarify* with candidates, or *refuse* with a short reason. A question is labeled *answer* when exactly one commune-level procedure (or a pair of equivalent ones) is clearly what the citizen needs; *clarify* when several different procedures fit, or when the procedure exists only in province-specific versions and no province is given; *refuse* when the procedure is not in the data or not handled at commune level, or the question concerns a submitted file, a complaint or petition, law and penalties, police internal matters, or no procedure.

Procedure data. The National Public Service Portal[1] was collected once on 13 October 2026 through its public, unauthenticated JSON endpoints (it has no official API) at two requests per second: 1,517 procedures, of which the 542 commune-level procedures that are not province-specific form the scope.

Real questions. We pooled two public Hugging Face datasets of citizen questions sent to the Binh Dinh provincial portal[19] and the Ministry of Public Security portal[20], masked personal data (patterns for phone and identity numbers, e-mail addresses and introduced names, plus a manual list), and removed duplicates, lines under four words, greetings and 266 questions used earlier to design clarification rules, leaving 1,490 questions. Samples were drawn with fixed seeds from a *random* stratum and a *topical* stratum (keywords of commune-level fields, fixed before any system output was seen, to obtain more answerable questions):

- **development set**: 282 questions (55 answer, 7 clarify, 220 refuse);
- **first test set**: 313 other questions (59 answer, 6 clarify, 248 refuse; 88% refuse in its random stratum), locked before any code change; 11 are near-duplicates of development questions (Sec. 6.2);
- **second test set**: the rest of the pool minus 25 near-duplicates, random stratum only, in an order locked before the second improvement round. The project lead (an author) labeled the first 400 during that round, knowing the kinds of rules being added but neither the rules nor any output on these questions; Gemini 4 Argon proposed each label, checked item by item, under the development guideline with two clarifications added (petitions are refused rather than set aside; two near-equivalent procedures make one *answer*).

  After removing 37 items that the guideline sets aside as truncated or unintelligible, 363 remain (21 answer, 7 clarify, 335 refuse: 92%).

Labels and agreement. Development and first-test labels were produced, following a written guideline, with the same AI assistant that wrote the rules (for development, two separate passes by that assistant, with disagreements resolved as it proposed and approved by the project lead). For 100 random first-test questions, Gemini 4 Argon proposed new labels and the project lead checked each without seeing the first pass, though having seen the aggregate class proportions. On the 91 questions that both passes kept (the second set aside nine petitions as non-questions, which led to the first clarification above), they agree on 79, with Cohen's kappa[21] κ=0.685 (95% bootstrap interval 0.55–0.82); 8 of the 12 disagreements are *answer* versus *clarify* between related procedures. The first-pass labels were not changed.

Team-written set. For comparison we use the blind part of the team's regression set: 518 questions written over several rounds, 102 imitating citizens, with 384 *answer* (381 whose procedure is still in the data), 76 *clarify* and 58 *refuse*.

4. System and Variants

Base system. The assistant as it stood before this study uses no language model to decide. Each segment of a question is normalized (abbreviations, synonyms, requested fields such as fees) and procedures are ranked by inverse document frequency over whole Vietnamese syllables, with a penalty for mismatched diacritics and thresholds on score and coverage (the share of the question's weight found in the name). A policy module checks the plan against the data and outputs the behavior; answers are composed by code from the record, and an optional LLM rewriting step is off in all experiments.

Rule layer (A). Rules written from the base system's errors on development, each behind a switch (all off reproduces the base output on every test question): *letter stripping* removes greetings, the addressed agency and thanks, which otherwise match procedure names; a *citizen lexicon* of about twenty patterns maps citizen wording to current official names, including terms changed by recent laws; a *weak-name filter* stops a few shared words from naming a long procedure; an *out-of-scope gate* refuses questions about one's own file, complaints and petitions, law and penalties, police internal matters and topics absent from the data, unless they ask an explicit procedural question; *focus* keeps only the segments of a long letter that ask something; and an *organization default* prefers the individual version of a procedure.

Second round (A2). Written after the first test set had been run, on the remaining errors of A on the development and first test sets, behind one more switch: a lexicon that tolerates inserted words; an explicit request recognized by the lexicon overrides a procedure inferred from a life event; more out-of-scope patterns (plain-word status questions, private disputes); and, as the labeling guideline prescribes, the version common to all provinces is answered instead of asking for the province.

Reranker (B). The rule ranker's top eight and the ten in-scope procedures closest under BGE-M3 embeddings[22] are scored by the cross-encoder bge-reranker-v2-m3[23]; the system refuses below a score threshold and replaces the rule's choice when another candidate wins by a margin (thresholds chosen by grid search on development).

LLM reader (C). A local 4-billion-parameter model (Qwen3-4B-Instruct-2507, Q4_K_M, Ollama, temperature 0, JSON schema)[24,25] classifies the intent (procedure, file status, complaint, law, internal, other) and, for procedure questions, names the procedure in words, which the rule ranker maps to a procedure; other intents become refusals, and the *name-only* variant ignores the intent.

Learned gates (L). Trained on the development labels (282 questions for the first test set, 595 for the second) to predict *refuse*, a classifier replaces the hand-written gate on top of the base system: multinomial naive Bayes over word and character n-grams, or class-balanced logistic regression over BGE-M3 embeddings of the question, with threshold and regularization chosen by five-fold cross-validation. Both were added after the test sets had been run, as trained counterparts of the gate.

RAG baseline (R). In a common design in which the model decides, BGE-M3 retrieves the five closest in-scope procedures and the same 4B model returns *answer* with a number, *clarify* or *refuse*, without any rule; its prompt lists

the refusal categories of the labeling guideline. After a first, short prompt over-used clarification, two variants were declared together; the best on development (ordered decision rules: 0.445, against 0.343 and 0.432) was run once on each test set.

5. Evaluation Protocol

Order of work. Each test set was labeled and locked by SHA-256 and a commit; rules and thresholds were developed on the development set only; the code was frozen in a commit; each configuration was run once and its results read once. Reusing a test set for selection inflates its estimates[26], so no test set was used to choose anything. The first test set then joined the development set (595 questions) for the second round. Ablations and the learned gates were run afterwards, on the frozen code, as *post hoc* analyses.

Selection and comparisons. Written before each test run: the development configuration with the highest balanced accuracy, subject to end-to-end accuracy not lower than that of the rule layer alone, ties going to end-to-end accuracy. This selected A + B in the first round and A2 + B in the second (0.916, tied with A2 but with one more end-to-end answer); such development numbers are optimistic, as the rules were written on the same questions. The first plan compared the base with each variant and with the selection; the second declared three primary comparisons, A + B → A2 + B, base → A2 + B and A2 + B → R. All other comparisons are descriptive.

Metrics. *Behavior accuracy* is the share of correct three-way decisions; since always refusing scores 79.2% and 92.3% on the two test sets, the main metric is *balanced accuracy*, the mean recall of *answer* and *refuse*[27] (*clarify* is too rare to weigh equally). Because it counts an answer with the wrong procedure as correct, we also report *end-to-end* accuracy, the *answer* questions answered with a correct procedure first, and answer *precision*, the share of all answers given that are correct; *top-1* accuracy is the share of *answer* questions whose first output procedure is correct (a refusal outputs none). Proportions carry Wilson 95% intervals[28]; paired comparisons use the exact two-sided McNemar test[29] and, for balanced accuracy, a paired bootstrap 95% interval of the difference (4,000 resamples).

Setup. Procedure data of 13 October 2026; one laptop with an NVIDIA GeForce RTX 4050 (6 GB), paced by a temperature guard.

6. Results

6.1 Team-written versus real questions

On the same frozen base system, behavior accuracy is 77.2% (95% interval 73–81%) on the 518 team-written questions but 56.0% (50–62%) and 56.5% (51–62%) on the real development and first test sets; balanced accuracy falls from 0.852 to 0.636 and 0.647, and top-1 accuracy on *answer* questions from 274/381 (71.9%) to 7/55 (12.7%) and 20/59 (33.9%) (5/42 against 17/48 in the topical stratum, a gap we cannot explain). The largest change is in what is asked: 78 –79% of real questions should be refused, against 11%, and the base system refuses about half of them (116/220 and 132/248, against 52/58 team-written). On the team-written set the rule layer makes no net change in behavior (+4/−4; 400/518 in both cases), so a team-written set alone would have shown neither the problem nor the fix.

6.2 Locked test sets

First test set. The rule layer (Table 1) raises balanced accuracy from 0.647 to 0.814 (difference +0.167, 95% interval 0.095–0.240) and end-to-end accuracy from 19/59 (95% interval 22–45%) to 35/59 (47–71%): base → A gains 96 and loses 10 questions on behavior ($p \approx 9 \cdot 10^{-19}$) and 22 against 6 end-to-end ($p=0.004$). Correct refusals rise from 132 to 219 of 248 while correct *answer* decisions stay level (45 to 44). The gain holds without the 11 near-duplicates (0.651 to 0.810), in the random stratum (102 questions, 11 *answer*; 0.512 to 0.859) and in each portal (Public Security 0.618 to 0.875, Binh Dinh 0.649 to 0.751). In *post hoc* ablations, removing the gate costs 70 correct refusals but adds 4 end-to-end answers, the lexicon costs 17 end-to-end answers and focus 7; the other rules change at most one question. Alone,

the reranker (0.692) and the LLM reader (0.785; against A +28/−35 on behavior, p=0.45, and 25 against 35 end-to-end, p=0.06) do not do better, nor does either learned gate differ significantly in balanced accuracy from the hand-written gate alone on top of the base system (naive Bayes 0.765, embeddings 0.801, against 0.770). Added to the rules, the reranker gains one question; the reader's intents add refusals (+25/−10, p=0.017) but lose answers, and its name alone scores highest (0.834) but was not selected and gains nothing significant end-to-end (+8/−4, p=0.39). The RAG baseline scores 0.489, no better than always refusing, as it asks for clarification on 118 of the 248 questions it should refuse (0.727 if these were refusals, still below A). No configuration recognizes more than one of the six *clarify* questions, and second-pass labels give the same picture (base → A on 91 questions: +24/−1).

Second test set. On new questions the rule layer again changes refusals most: correct refusals rise from 175 to 297 of 335 and balanced accuracy from 0.571 to 0.705 (+125/−5, $p \approx 4 \cdot 10^{-31}$; difference +0.134, 0.033–0.228), the gate again supplying the refusals (62 fewer without it, 2 more end-to-end answers) and the lexicon the answers. The learned gates refuse more than the hand-written gate alone (308 and 313 against 287) but decide *answer* on only 5 of the 21 answerable questions against 10, for lower but not significantly lower balanced accuracy (0.579 and 0.586 against 0.666). The second round does not carry over on behavior: in the comparison declared before the run, A2 + B, which had gained 25 questions and lost none against A on development, ties A + B (+3/−3; end-to-end 10 against 8 of 21, p=0.63; balanced difference +0.046, −0.048 to 0.143, a wide interval, since only 21 questions are answerable) and A (+4/−4) while beating the base (+129/−9). R is again last (0.413; 0.574 with clarifications as refusals; against A2 + B +19/−122, $p \approx 10^{-19}$).

Answer precision. Answers remain unreliable: of the questions A answers, 45% get a correct procedure on the first test set and 20% on the second, most wrong answers going to questions that should be refused (29 of 42 and 34 of 37). On random traffic no configuration answers correctly more often than wrongly: the most precise are A + C with 7 of 15 answers in the random stratum of the first test set and A2 + B with 10 of 50 on the second.

Table 1. Locked test sets, one run each (313 and 363 questions; Sec. 3). Bal.: balanced accuracy; Ref.: correct refusals; E2E: end-to-end correct answers (of 59 and of 21); Prec.: share of given answers that are correct (%); Gate only: the out-of-scope gate of A alone on the base system; L: learned gates in its place; **post hoc*; –: not run on that set (for A2 + B on test 1: A2 was developed on those questions).

	Test 1				Test 2			
Configuration	**Bal.**	**Ref.**	**E2E**	**Prec.**	**Bal.**	**Ref.**	**E2E**	**Prec.**
Always refuse	0.500	248	0	–	0.500	335	0	–
Base	0.647	132	19	11.7	0.571	175	5	3.0
B: reranker alone	0.692	175	21	17.9	–	–	–	–
C: LLM reader alone	0.785	213	25	30.9	–	–	–	–
R: RAG + LLM decides	0.489	104	28	43.8	0.413	197	4	10.5
L: naive Bayes gate*	0.765	224	14	21.9	0.579	308	2	6.9
L: embedding gate*	0.801	225	17	25.4	0.586	313	2	8.0
Gate only*	0.770	218	15	20.8	0.666	287	5	9.1
A: rule layer	0.814	219	35	45.5	0.705	297	9	19.6
A + C (intent, name)	0.817	237	32	58.2	–	–	–	–
A + C (name only)	0.834	216	39	47.0	–	–	–	–
A + B (round-1 selection)	0.823	219	36	46.2	0.683	298	8	17.4
A2 + B (round-2 selection)	–	–	–	–	0.729	297	10	20.0

7. Discussion

What the rules do. The out-of-scope gate supplies the refusals: out-of-scope questions fall into a few recurring kinds (file status, complaints, penalties, police internal matters) worded regularly enough for patterns, and classifiers trained on the same labels were not significantly better in balanced or end-to-end accuracy on either test set. The citizen lexicon supplies the answers by mapping citizen wording to official names. The reranker favors complaints that name a real procedure, and the LLM reader labels many long procedural narratives as complaints.

What remains. Of the 24 *answer* questions that A misses on the first test set (*post hoc* analysis), 12 are refused, 4 of them by the gate although the base system had ranked the correct procedure first; of the other 12, 7 have the correct procedure outside the top five, typically in long letters that only imply it. With such low answer precision on random traffic, an assistant should have a human officer confirm its answers, or answer only when confident[11]; refusing better is necessary but not sufficient.

Why the second round did not carry over. A2 + B changes the decision or procedure of a similar share of questions on development and on the new set (6.2% and 5.2%), but with 30 gains and no losses on development against 4 and 4 on the new set (a decision counting as correct only with the right procedure); its new out-of-scope patterns fixed 15 wrong answers on development but one on the new set, which suggests that they fit particular phrasings. The small *answer* class and labels from another model with a slightly extended guideline may also contribute.

Limitations and threats to validity. (i) One domain, language and data snapshot: hand-written rules for Vietnamese commune-level procedures, with no claim about provincial or ministry procedures. (ii) With only 59 and 21 *answer* and 6 and 7 *clarify* test questions, small differences are not significant and nothing can be concluded about clarification. (iii) The development and first-test labels were produced with the AI assistant that wrote the rules, so the gate may encode its notion of scope; the second-pass sample and the second test set were labeled with another model and checked by the project lead, who had seen aggregate class proportions and knew the kinds of rules being added. Circularity alone does not explain the second test set, whose labels did not come from that assistant (though they follow the same guideline) yet where the rule layer still gains 125 questions and loses 5 on behavior (122 more correct refusals). No fully independent human annotation exists, however. (iv) All questions come from two portals, with a larger gain on one, and the 79% refusals of our samples understate the share in random traffic. (v) The locks are commits in our own repository, without an external timestamp. (vi) The rules received far more development effort than the learned components and the RAG baseline, systems from other studies could not be run on our questions, and the ablations and learned gates are *post hoc*. (vii) LLM components ran once at temperature 0 on one quantized model.

8. Conclusion

For an assistant limited to commune-level procedures, the hard part of real citizen questions is deciding whether to answer at all. A system that looked strong on team-written questions answered 114 of the 248 real questions it should have refused. A small, transparent rule layer nearly doubled the end-to-end correct answers on a locked test set and kept its gain in refusals on a second one; neither a reranker nor a local 4B LLM reader significantly improved balanced or end-to-end accuracy, nor did two gates learned *post hoc* in place of the hand-written one, and letting the model decide was no better than always refusing. A second round of rules that looked better on development tied the first on new questions, and answers remain wrong more often than right on random traffic. Future work: confidence-based answering with human confirmation, procedure names proposed by the model while rules decide, a third locked set, and independent human annotation.

Ethics and Data Availability

The procedure records are public, were collected without logging in (Sec. 3) and contain no personal data. The citizen questions come from public datasets whose cards state no license; personal data were masked before labeling, no names or identifiers appear in this paper, and no data were collected from human participants, so no ethics approval was required. We use them only for evaluation and do not redistribute the question texts. The code, the locked question

identifiers and hashes, the labels and the per-question decisions of every run are at https://github.com/3-hWnG/intern-amazing-group_7/tree/paper1-vjcs-draft; each identifier maps to a row of the original datasets, and the full commit history is available to reviewers on request.

Competing Interests

The authors declare no competing interests.

Declaration of AI Use

During the preparation of this work, the authors used Claude Opus 5.5 (Anthropic) to assist with writing and modifying the software, running and summarizing the experiments, first-pass annotation of the development and first test sets (Sec. 3), and drafting and language editing of the manuscript; the code that builds the procedure database was first written with Claude Opus 5 and later modified with Claude Opus 5.5. The same assistant that produced the first-pass labels also wrote the rules evaluated here. Gemini 4 Argon (Google) proposed the labels of the second-pass sample and of the second test set, which the project lead (the first author) reviewed item by item. The research questions, the evaluation protocol and the final decisions were made by the authors, who reviewed, verified and edited the output and take full responsibility for the content of the publication.

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